MATH 1271 **Name:** _____
Fall 2026
Quiz 12: 10/22 Content
10/27/2026
Time Limit: 5 minutes **Student ID:** _____

This quiz consists of 1 questions across 2 pages worth 5 points.
Rules:

- You have the indicated amount of time above to complete the quiz and you must not begin working on the quiz until instructed to do so.
- You must show all of your work to receive full credit, unless stated otherwise. **On problems that require work to be shown, answers that are mysterious or unsupported will not receive full credit.**
- You may not use a calculator, textbook, notes or any other external resources.
- Please write your initials at the top of every page in the event the pages become separated.
- If you run out of space on a page, feel free to use the back of the page to continue your work.
- You understand that if you violate any of these or the University's rules, you will receive a score of zero on this quiz and you may be subject to further disciplinary action.

Following the University of Minnesota's policy on academic honesty, I pledge that the work shown on this quiz is entirely my own and that I have followed the rules stated above.

Signature: _____

Question	Points
1	5
Total:	5

1. (5 points) Oil spilled from a ruptured tanker spreads in a circle whose area increases at a constant rate of $6\text{mi}^2/\text{hr}$. How rapidly is the radius of the spill increasing when the area is 5mi^2?

Solution: The area of a circle in terms of its radius is given by

$$A = \pi r^2. \tag{1}$$

We differentiate this formula with respect to time:

$$\frac{d}{dt}A = \frac{d}{dt}\pi r^2 \tag{2}$$

$$\frac{dA}{dt}\frac{d}{dA}A = \pi\frac{dr}{dt}\frac{d}{dr}r^2 \tag{3}$$

$$\frac{dA}{dt} = 2\pi r\frac{dr}{dt}. \tag{4}$$

So when $\frac{dA}{dt} = 6\text{mi}^2/\text{hr}$ and $A = 5\text{mi}^2$, we see that $5 = \pi r^2$, so $r = \sqrt{\frac{5}{\pi}}$. Therefore at that time:

$$\frac{dA}{dt} = 2\pi r\frac{dr}{dt} \tag{5}$$

$$6 = 2\pi\sqrt{\frac{5}{\pi}}\frac{dr}{dt} \tag{6}$$

$$3 = \sqrt{5\pi}\frac{dr}{dt} \tag{7}$$

$$\frac{dr}{dt} = \frac{3}{\sqrt{5\pi}}\text{mi/hr} \tag{8}$$

So the radius of the spill is increasing at a rate of $\frac{3}{\sqrt{5\pi}}$ miles per hour.